

Duplicate MRN Analysis Project

Using Mockaroo, SQL, and Power BI to uncover
the root causes of duplicate patient records

Understanding the Business Problem

What is an MRN?

Duplicate Medical Record Numbers (MRNs) occur when a single patient is assigned multiple unique identifying numbers within a healthcare system, often due to data entry errors, name changes, or system mergers. These duplicates cause critical safety risks, including missed allergies, incorrect medications, and fragmented care histories.

Why is this important?

- **Patient Safety:** Inaccurate medical records can result in hidden or conflicting information
- **Cost Impact:** Redundant testing and billing errors
- **Operational Drag:** Staff spend significant time manually identifying and interpreting duplicate records instead of higher-value work

What can be done about this?

Data analysts can create visuals with collected data to reveal and analyze trends which can help with identifying the lead causes. These visuals can be created with platforms such as Power BI or Tableau, which can aid management in making informed business decisions.

Project Objectives

Which facilities and service lines generate the most duplicate MRNs?

Which patient demographics are most affected?

What are the root causes driving duplication?

Workflow

Generate synthetic data

10,000 rows across 19 columns of data were synthetically generated to simulate a realistic Centralized Healthcare Database (CHD), using Mockaroo. Constraints written in Ruby were added to provide realistic data, such as skewing over 60% of patients in the "Senior" age group.

DAX Formulas

Measures were written and compiled into a "_DAX" table, which centralizes all DAX formulas for better organization.

Creating Visuals and Polishing for Aesthetics

Visuals were created using DAX measures and columns in the 3 tables. A final polish was done to ensure that the report is easy to read, and that the styling theme was consistent throughout the entire report. A PDF version of the report was exported, and was converted into separate PNG files for convenience.

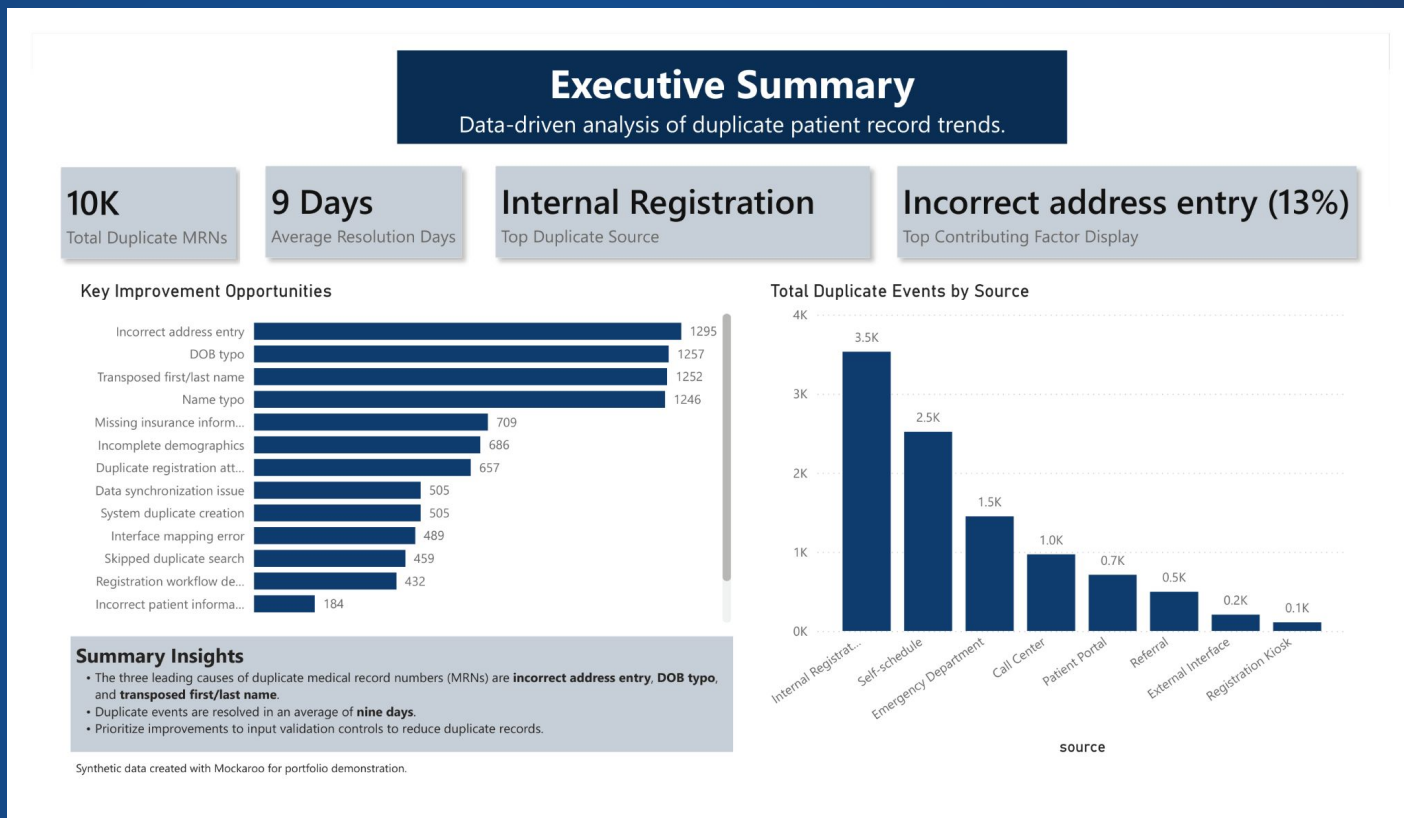
Organize and Clean Data

The main fact table was divided into 3 tables to avoid the use of unnecessary duplicate values. It was structured into a star schema, consisting of 1 fact table and 2 dimension tables.

Planning Stage

Each report page was given a name, purpose, and defined scope, with Power BI used to visualize the underlying data.

- The 'Executive Summary' report page provides a high-level snapshot of duplicate MRN concentration
- Sets the stage for deeper investigation into root causes



- The 'Root Cause Analysis' page shows a detailed view of underlying causes
- Helps identify the location and source of the causes
- Establishes the average duplicate MRNs per facility as a benchmark for comparison against the highest-volume facilities.

Root Cause Analysis

- 63% of duplicate MRNs are **adult patients**.
- Ambulatory accounts for the largest share.
- Montefiore Medical Center has the highest duplicate MRN volume of any facility, exceeding the national average by roughly **47 records**.

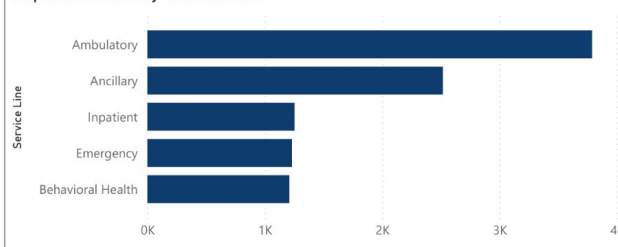
1.78

Avg Duplicate MRNs per Facility

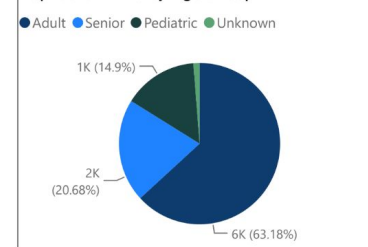
Root Cause Analysis

Pinpointing where and why duplicate MRNs originate.

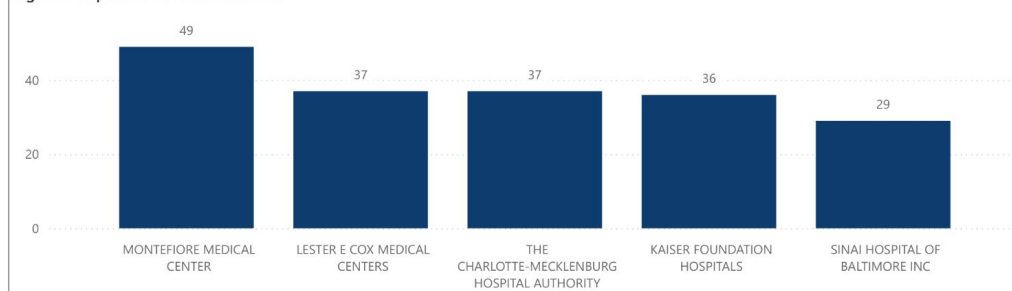
Duplicate MRNs by Service Line



Duplicate MRNs by Age Group



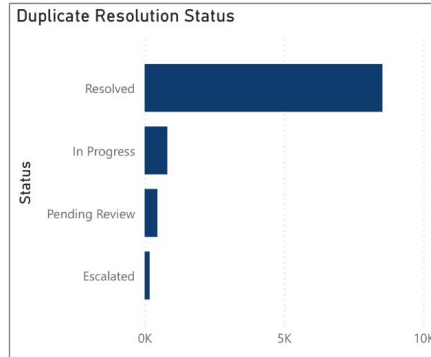
Highest Duplicate Volume Facilities



- Provides a benchmark % resolved rate to gauge overall process performance.
- Compares investigation time against resolution time by department to flag efficiency gaps

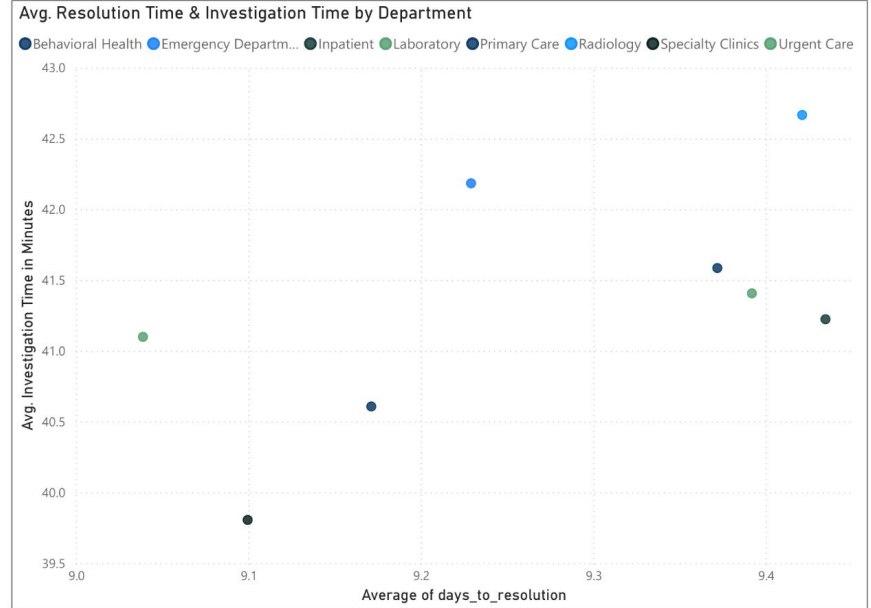
Resolution Tracking

Evaluating resolution speed and success across departments.



0.85

% of Duplicate MRNs Resolved



Actionable Recommendations

Prioritize High-Volume Facilities

Focus process improvements on outliers like Montefiore Medical Center

Invest in automated patient-matching software

Flag likely duplicate patients in Ambulatory and Ancillary before a new MRN is even created

Set department-level resolution targets

Establish time-to-resolve benchmarks for slower departments and reallocate staff or tools to close the gap

Improve Input Validation

Implement stricter data entry controls and real-time duplicate-checking at registration to catch inconsistencies (name variations, DOB mismatches, address changes) before they generate a new MRN

Questions / Thank You

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